

Supplementary material from "An Entropy-based Framework for Hybrid Coalitions in Game Theory. Part I: Human arbitration"

Chapter I: from $H = 200$ to $H = 5,000$

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1 Introduction

This supplementary material accompanies the revised version of “An Entropy-Based Framework for Hybrid Coalitions in Game Theory—Part I: Human Arbitration.” All theoretical definitions, delegation rules, learning-rate specifications, policy-update conventions, and equilibrium concepts are those established in the main paper and are not repeated here [1].

The purpose of this chapter is to document the finite-horizon computational evidence supporting the Human–AI policy-alignment mechanism developed in the paper. The analysis covers the independently executed horizons $H \in \{200, 1000, 5000\}$, which represent, respectively, the exploratory transient, early learning, and emerging-stabilization stages of the simulated dynamics.

For each horizon, the results are organized by the three policy-update schedules considered in the revised study—constant gain, global diminishing gain, and exact empirical updating—and by the two state-generation mechanisms: action-dependent endogenous transitions and exogenous state sampling. These combinations are reported separately because the learning schedule and the frequency with which each state is visited affect the finite-horizon path of policy adaptation.

The figures are organized around three complementary dimensions. All formal definitions and notation are those established in the main paper.

1. **Policy trajectories.** These plots show the Human and AI policy probabilities in states A and B . Most panels report action a_0 ; selected a_1 panels are additionally shown at $H = 200$ to illustrate the complementary structure of the binary policies.
2. **Utility-credit traces and value iterates.** These plots report the EWMA utility-credit traces and the finite-time computational Bellman iterates along the realized simulation path. They are used as numerical diagnostics of the implemented dynamics.
3. **Jensen–Shannon divergence and execution frequencies.** These plots report D_{JS}^T , the two delegation thresholds, and the cumulative Human and AI execution frequencies. They provide the most direct graphical evidence of policy alignment and the stabilization of execution authority.

In the policy panels, the horizontal-axis label *Decision instant T* identifies the interaction turn. The vertical-axis label *Predecision probability* refers to the policy available at that same instant after the state has been observed but before proposals, delegation, and execution. The subsequent effective update produces the policy available at $T + 1$.

Because the simulated action space is binary,

$$p_i(a_1 | s, T) = 1 - p_i(a_0 | s, T), \quad i \in \{H, AI\}.$$

Accordingly, the a_1 curves are derived complements of the corresponding a_0 curves rather than independently estimated policy components. This explains both the label *Derived predecision probability* used in the a_1 plots and the complementary symmetry highlighted for the selected constant-gain configurations at $H = 200$.

The simulations use the two state-generation mechanisms defined in the main paper [?]. When `r_s=False`, Virtual Nature follows the action-dependent Markov kernel $P(\cdot | s_T, a_T^*)$, so p_A and p_B are not applicable. When `r_s=True`, states are generated exogenously and independently of the executed action, with $\Pr(s_T = A) = p_A$ and $\Pr(s_T = B) = 1 - p_A$, where p_A is supplied by the simulation grid.

2 Early exploratory behaviour: $H = 200$

2.1 Policy trajectories

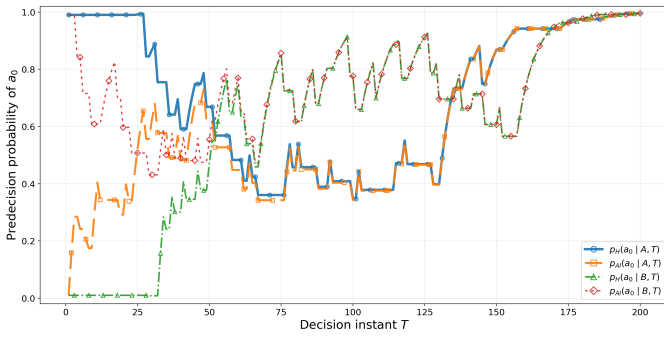
For the state-wise comparisons in this subsection, define

$$D_{JS,\max}^{\text{final}} = \max\{D_{JS,A}^{\text{final}}, D_{JS,B}^{\text{final}}\}.$$

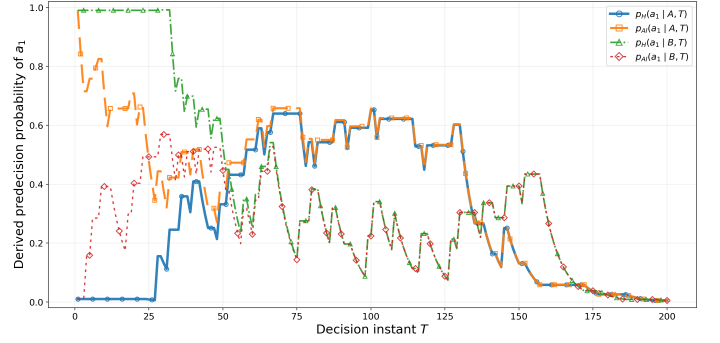
This diagnostic prevents a negligible divergence in the terminally observed state from being interpreted as alignment in both states. Because the action space is binary, the displayed a_1 trajectories are the pointwise complements of the corresponding a_0 trajectories.

2.1.1 Constant- η learning

Endogenous state generation. With $\eta = 0.15$ and $(\alpha_{\text{agree}}, \alpha_{\text{disagree}}) = (0.35, 0.85)$, the action-dependent transition kernel produces substantial coverage of both states: 41.5% of the decisions occur in A and 58.5% in B . The initially irregular trajectories contract until the Human and AI policies are nearly identical in both states. The final a_0 probabilities are approximately 0.9949 in A and 0.9961 in B , with $D_{JS,\max}^{\text{final}} = 1.03 \times 10^{-9}$. The AI executes 97.0% of the decisions, while agreement occupies 88.5% of the realized path. Thus, the visible oscillations are compatible with strong finite-horizon alignment when both state-conditioned policies receive repeated updates.



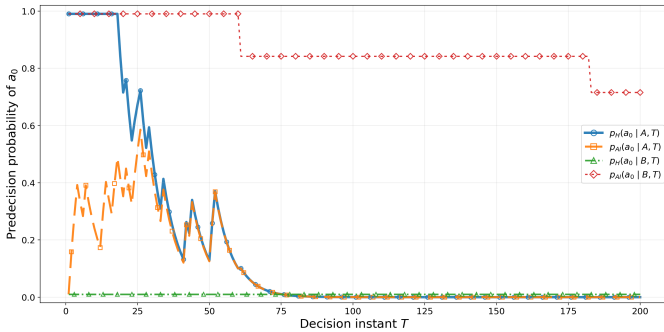
(a) Predecision probability of a_0 ($r_s = F$).



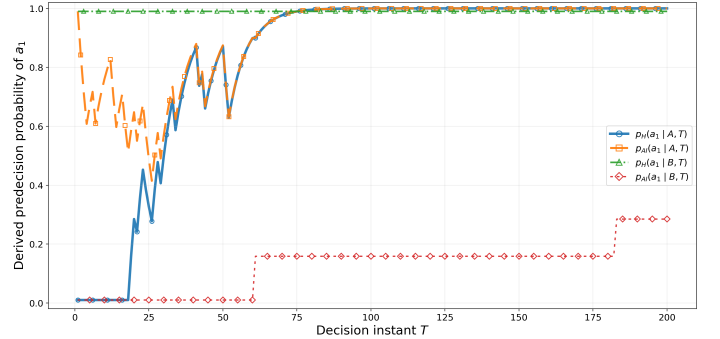
(b) Derived predecision probability of a_1 ($r_s = F$).

Figure 1: Binary-action policy symmetry at $H = 200$ under constant- η learning with $\eta = 0.15$ and endogenous state generation. The right-hand trajectories are the pointwise complements of the corresponding left-hand trajectories.

Exogenous state generation and rare-state coverage. With $r_s = T$ and $p_A = 0.99$, state B is visited only twice. The frequently sampled state A becomes aligned, but the final Human and AI probabilities of a_0 in B remain 0.0100 and 0.7153, respectively, giving $D_{JS,B}^{\text{final}} = 0.4735$. Although the AI executes 97.5% of the decisions and the agreement frequency is 90.5%, these frequencies are dominated by state A and do not establish two-state alignment. Across the complete constant- η grid, practical alignment is reached in 67.50% of the endogenous cases but only 23.33% of the exogenous cases; the extreme values $p_A \in \{0.01, 0.99\}$ produce no practically aligned exogenous configuration at this horizon.



(a) Predecision probability of a_0 ($r_s = T$, $p_A = 0.99$).



(b) Derived predecision probability of a_1 ($r_s = T$, $p_A = 0.99$).

Figure 2: Binary-action policy symmetry at $H = 200$ under constant- η learning with $\eta = 0.15$ and exogenous state generation. The extreme value $p_A = 0.99$ makes the lower-frequency state a diagnostic of insufficient state coverage.

2.1.2 Global diminishing- η learning

Endogenous state generation. The selected schedule is $\eta_T = 0.15/(1 + 0.0005T)$. Over $H = 200$, the learning rate falls only from 0.15 to approximately 0.136, so the trajectory remains close to the constant- η case. State frequencies are 42.5% in A and 57.5% in B , and the policies finish aligned at approximately 0.9559 and 0.8358 for action a_0 in states A and B , respectively. The maximum terminal state-wise divergence is 2.55×10^{-10} . AI execution reaches 97.0%, agreement reaches 88.5%, and the dominant finite-horizon influence is again state coverage rather than the still moderate reduction in η_T .

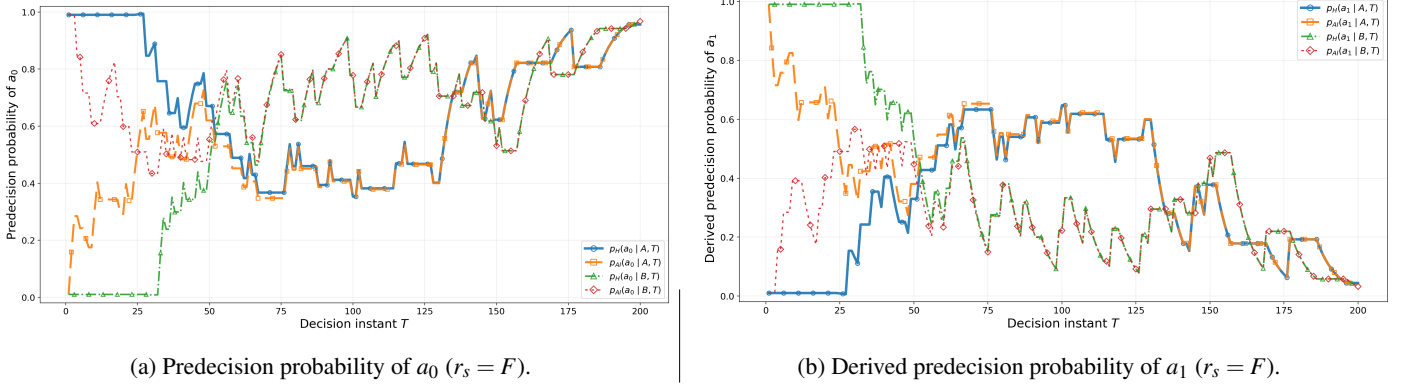


Figure 3: Binary-action policy symmetry at $H = 200$ under global diminishing- η learning with $\eta_T = 0.15/(1 + 0.0005T)$ and endogenous state generation.

Exogenous state generation and persistent rare-state misalignment. Under $p_A = 0.99$, state B is again visited only twice. Its final Human and AI probabilities of a_0 are 0.0100 and 0.7295, respectively, yielding $D_{JS,B}^{\text{final}} = 0.4890$, despite AI execution of 97.5% and agreement of 90.5%. In the full global-diminishing grid, practical two-state alignment is obtained in 62.50% of the endogenous configurations and 21.11% of the exogenous configurations. As under constant gain, no exogenous case with $p_A \in \{0.01, 0.99\}$ satisfies the practical criterion.

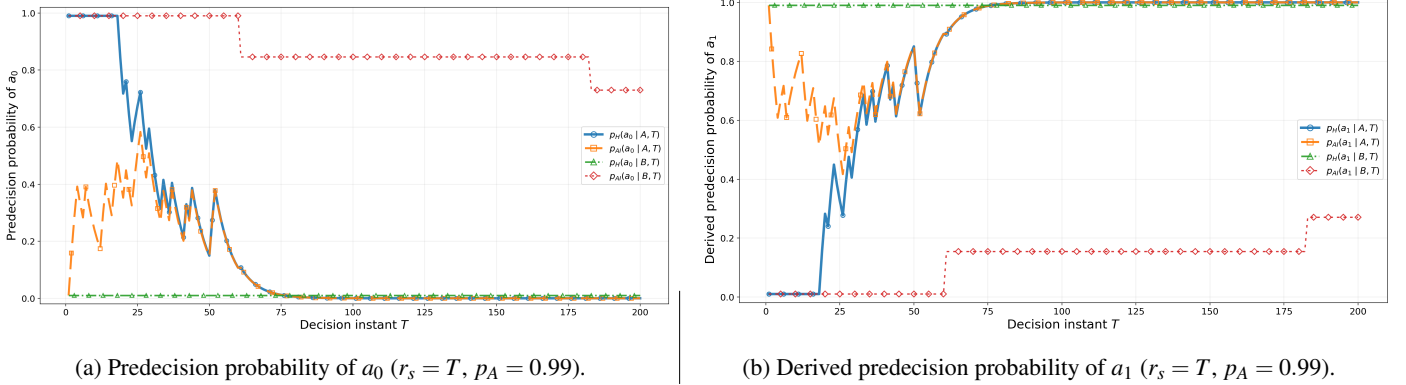
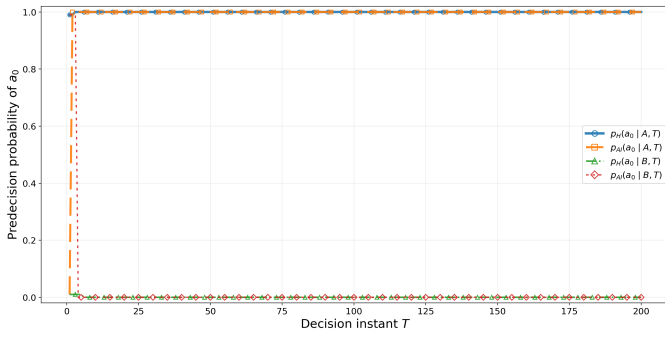


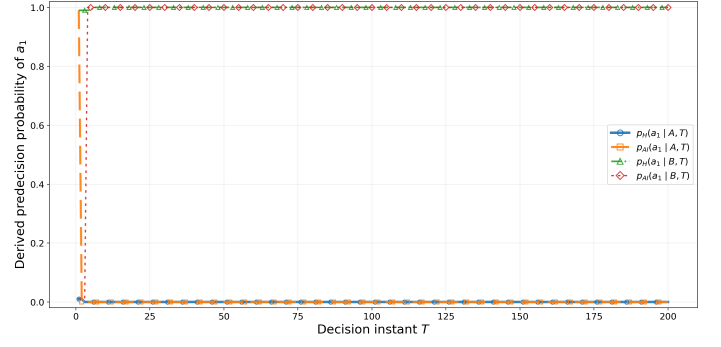
Figure 4: Binary-action policy symmetry at $H = 200$ under global diminishing- η learning with $\eta_T = 0.15/(1 + 0.0005T)$ and exogenous state generation. The rare-state trajectories remain insufficiently updated at $p_A = 0.99$.

2.1.3 Exact empirical learning

Endogenous state generation and immediate local averaging. Under exact empirical learning, $\eta_{i,T}(s) = 1/[N_i(s, T - 1) + 1]$, so the first effective update of an agent-state policy uses $\eta = 1$. With thresholds (0.20, 0.35), the two policies become identical at $p(a_0 | A) = 1$ and $p(a_0 | B) = 0$, with numerical alignment stable from approximately $T = 5$. State A accounts for 91.5% of the path and state B for 8.5%, but both receive effective observations. AI execution and agreement each reach 99.0%. The abrupt trajectories therefore document finite-sample lock-in under a local empirical average rather than gradual contraction.



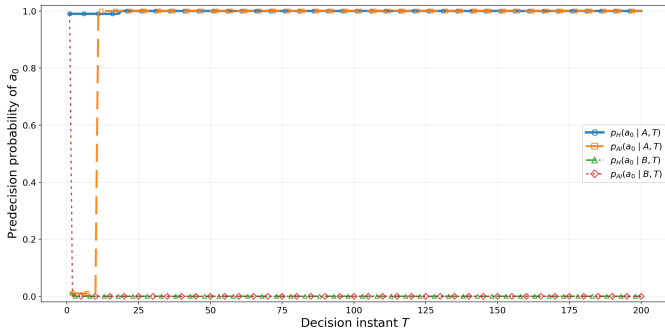
(a) Predecision probability of a_0 ($r_s = F$).



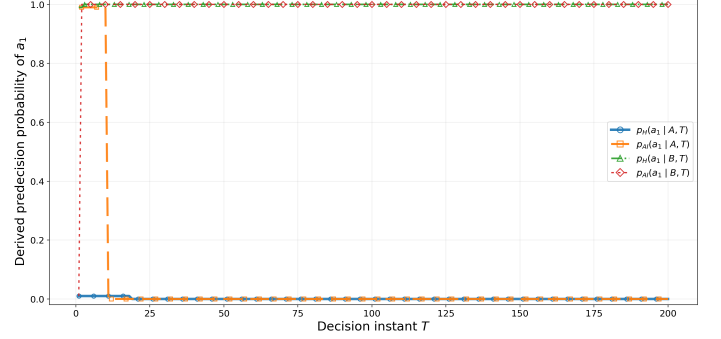
(b) Derived predecision probability of a_1 ($r_s = F$).

Figure 5: Binary-action policy symmetry at $H = 200$ under exact empirical learning with $\eta_{i,T}(s) = 1/[N_i(s, T-1) + 1]$ and endogenous state generation.

Exogenous state generation with repeated observations of both states. The exogenous configuration uses $p_A = 0.10$, producing realized state frequencies of 9.0% in A and 91.0% in B . Unlike the extreme $p_A = 0.99$ cases, both states receive repeated observations. Exact policy identity is established from approximately $T = 18$, with AI execution and agreement again equal to 99.0%. Every configuration in the complete exact-empirical grid satisfies both the practical and numerical state-wise alignment criteria. This result is specific to the local sample-average update and does not imply that all learning schedules converge uniformly at $H = 200$.



(a) Predecision probability of a_0 ($r_s = T$, $p_A = 0.10$).



(b) Derived predecision probability of a_1 ($r_s = T$, $p_A = 0.10$).

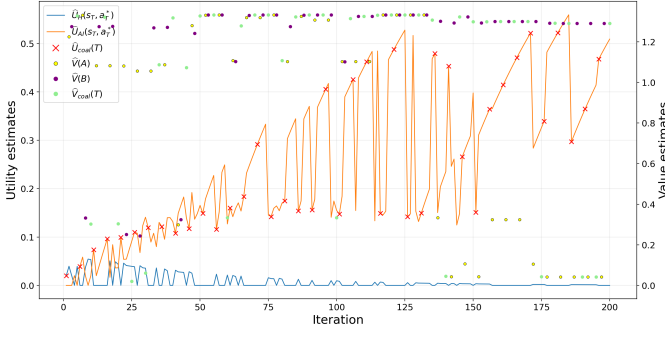
Figure 6: Binary-action policy symmetry at $H = 200$ under exact empirical learning with $\eta_{i,T}(s) = 1/[N_i(s, T-1) + 1]$ and exogenous state generation with $p_A = 0.10$.

2.2 Utility-credit traces and value iterates

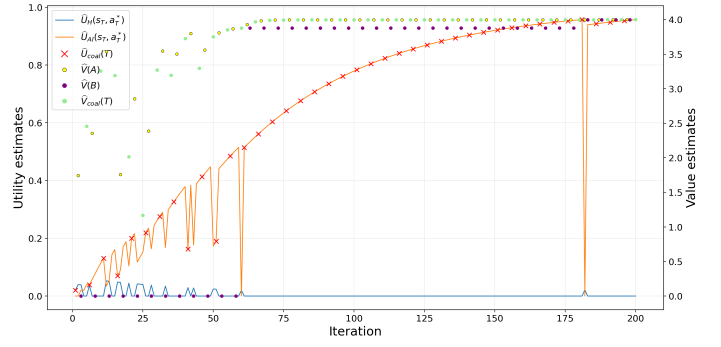
The displayed curves are secondary finite-time diagnostics of the realized interaction path. The quantities $\hat{U}_H$ and $\hat{U}_{AI}$ are EWMA utility-credit traces, $\hat{U}_{coal}$ is the credit trace associated with the owner of the executed proposal, and $\hat{V}^T$ denotes the asynchronous computational Bellman iterate. Abrupt changes may reflect switching among realized state-action entries rather than resets of the underlying tables. Because β and γ differ across configurations, vertical magnitudes are interpreted within each panel, not ranked directly across panels.

2.2.1 Constant- η credit and value dynamics

With endogenous state generation, the AI executes 97.0% of the decisions and agreement occupies 88.5% of the path. The Human credit becomes negligible, while the final AI and coalition credits coincide at 0.5093. The terminal value iterates are $\hat{V}(A) = 0.0417$ and $\hat{V}(B) = 1.2917$. The irregular path is consistent with repeated switching among states, actions, and owners during the exploratory transient. Under exogenous generation with $p_A = 0.99$, the trajectory is dominated by state A : AI execution reaches 97.5%, agreement reaches 90.5%, the final AI and coalition credits equal 0.9568, and the value iterates approach 4. The larger magnitude partly reflects $\gamma = 0.75$ and repeated visitation of the same state; it does not establish alignment in the state visited only twice.



(a) Endogenous generation ($r_s = F$); $\eta = 0.15$, $\beta = 0.02$, $\gamma = 0.25$.

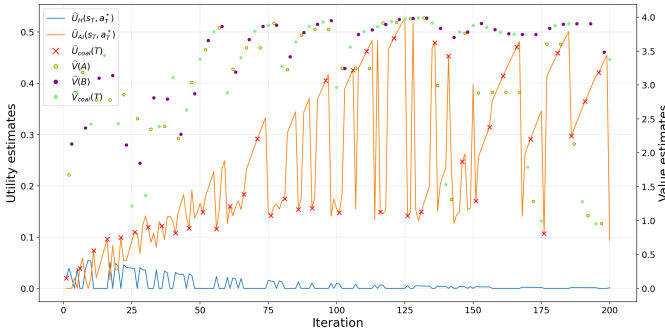


(b) Exogenous generation ($r_s = T$, $p_A = 0.99$); $\eta = 0.15$, $\beta = 0.02$, $\gamma = 0.75$.

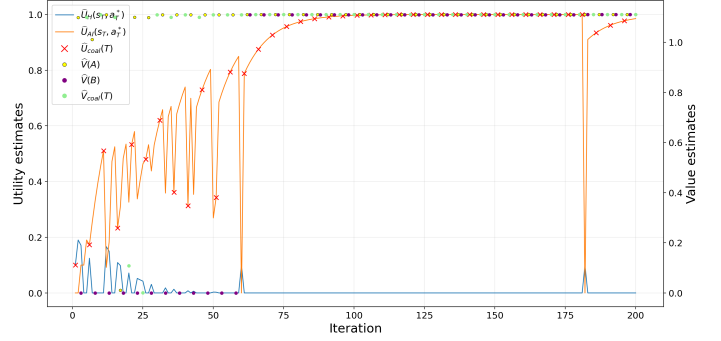
Figure 7: Utility-credit traces and finite-time value iterates at $H = 200$ under constant- η learning. The panels contrast balanced endogenous visitation with strongly imbalanced exogenous state coverage.

2.2.2 Global diminishing- η credit and value dynamics

The global schedule decreases from 0.15 to approximately 0.136 over the horizon, so its execution and regime frequencies remain close to those of constant gain. In the endogenous run, AI execution is 97.0%, agreement is 88.5%, the final AI and coalition credits are 0.0938, and the terminal values are $\hat{V}(A) = 0.9539$ and $\hat{V}(B) = 3.3821$. In the exogenous run with $p_A = 0.99$, AI execution reaches 97.5%, agreement 90.5%, the final AI and coalition credits reach 0.9850, and both value iterates are approximately 1.1111. Here $\beta = 0.10$ makes the exogenous credit trace more responsive, while $\gamma = 0.10$ limits the scale of the value recursion. As in the policy plots, the high agreement frequency is dominated by state A and does not remove rare-state misalignment.



(a) Endogenous generation ($r_s = F$); $\eta_T = 0.15/(1 + 0.0005T)$, $\beta = 0.02$, $\gamma = 0.75$.



(b) Exogenous generation ($r_s = T$, $p_A = 0.99$); $\eta_T = 0.15/(1 + 0.0005T)$, $\beta = 0.10$, $\gamma = 0.10$.

Figure 8: Utility-credit traces and finite-time value iterates at $H = 200$ under global diminishing- η learning. The small decay within this horizon leaves state coverage and the parameters β and γ as the most visible sources of contrast.

2.2.3 Exact empirical credit and value stabilization

Exact empirical updating generates a sharper transition. In the endogenous run, AI execution and agreement both reach 99.0%; the final AI and coalition credits are 0.8818, while the terminal values are numerically negligible. In the exogenous run with $p_A = 0.10$, both states receive repeated observations, AI execution and agreement again reach 99.0%, and the final AI and coalition credits equal 0.8585; the two value iterates end near zero. The rapid stabilization follows the local empirical schedule, whose first effective update of each agent-state component uses $\eta = 1$. These traces support the timing of the policy lock-in but remain secondary diagnostics: the grid ranking is determined by state-wise policy alignment, not by the final credit or value magnitudes.

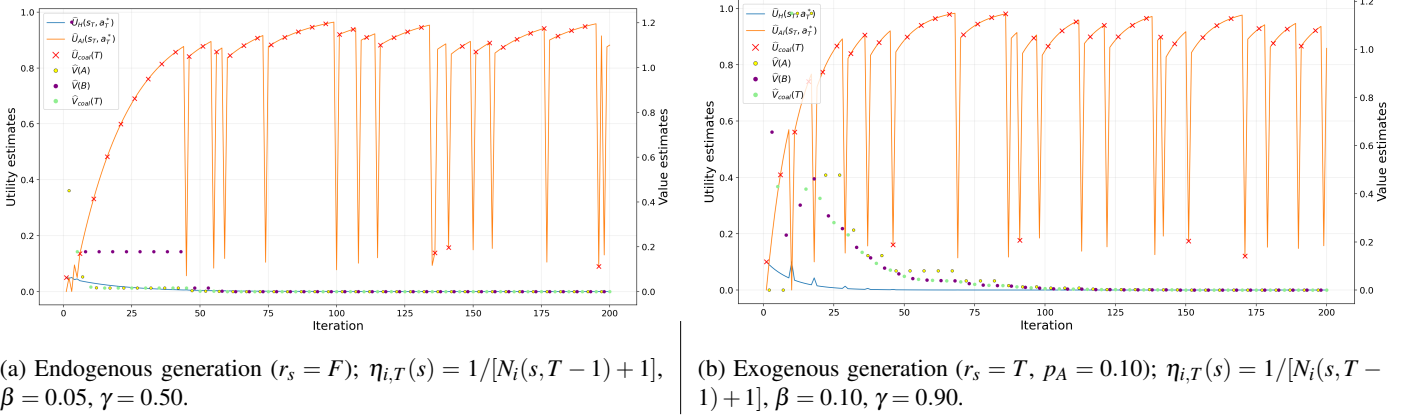


Figure 9: Utility-credit traces and finite-time value iterates at $H = 200$ under exact empirical learning. Both state-generation mechanisms display rapid stabilization after the first effective observations of the relevant state–action components.

2.3 Jensen–Shannon divergence and execution frequencies

The divergence curves report the predecision value D_{JS}^T in the state realized at decision instant T , together with the cumulative Human and AI execution frequencies. Consequently, a small terminal value of the displayed divergence does not by itself establish alignment in both states. The interpretation below also considers the state-conditioned terminal divergences and the empirical state visitation frequencies.

2.3.1 Divergence and execution under constant- η learning

Under endogenous state generation, the early trajectory crosses the three delegation regions before contracting below the agreement threshold. Practical two-state alignment becomes stable from approximately $T = 45$, and the terminal state-conditioned maximum is $D_{JS, \max}^{\text{final}} = 1.03 \times 10^{-9}$. The AI executes 97.0% of the decisions, while the Human executes 3.0%; agreement, contextual, and disagreement occupation frequencies are 88.5%, 10.5%, and 1.0%, respectively. The state frequencies, 41.5% for A and 58.5% for B, provide sufficient coverage for both policy components to align.

The exogenous configuration produces a superficially similar execution profile: AI execution reaches 97.5% and agreement occupies 90.5% of the horizon. However, $p_A = 0.99$ generates only two visits to state B. The isolated divergence spikes correspond to those rare-state observations. Although the divergence in the frequent state A becomes negligible, the terminal divergence in state B remains 0.4735. The panel therefore illustrates why cumulative agreement and AI-execution frequencies must be interpreted jointly with state-conditioned alignment and coverage.

i. Divergence and execution under constant- η learning

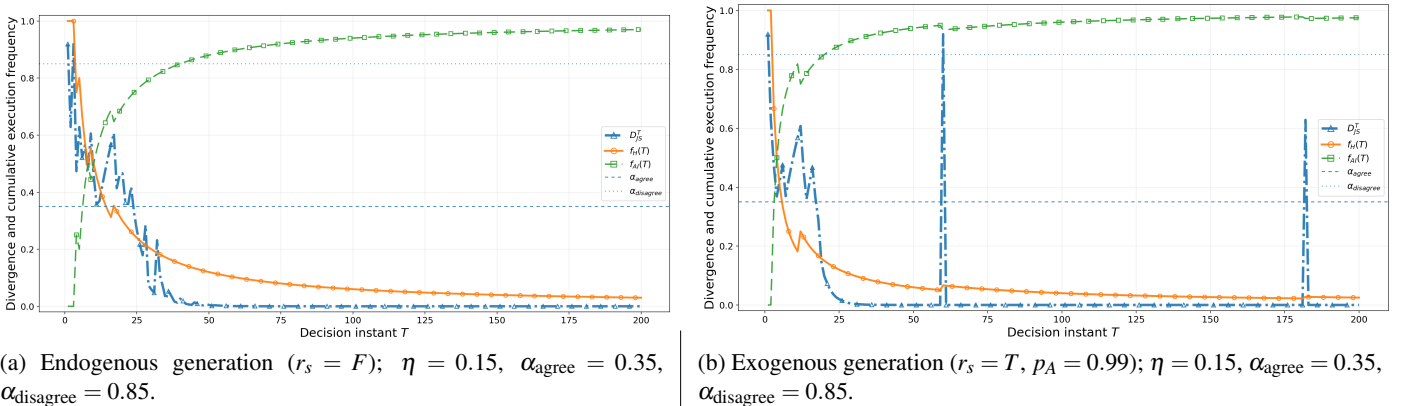


Figure 10: Jensen–Shannon divergence and cumulative execution frequencies at $H = 200$ under constant- η learning. Endogenous state coverage supports alignment in both states, whereas the exogenous case retains substantial divergence in the rarely visited state.

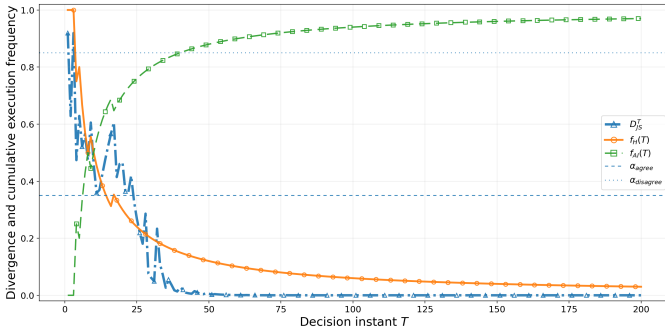
2.3.2 Divergence and execution under global diminishing- η learning

The global learning rate decreases only from 0.15 to approximately 0.136 over this horizon. Accordingly, the endogenous trajectory is close to the constant- η case: practical two-state alignment is stable from approximately $T = 46$, $D_{JS, \max}^{\text{final}} = 2.55 \times 10^{-10}$, and

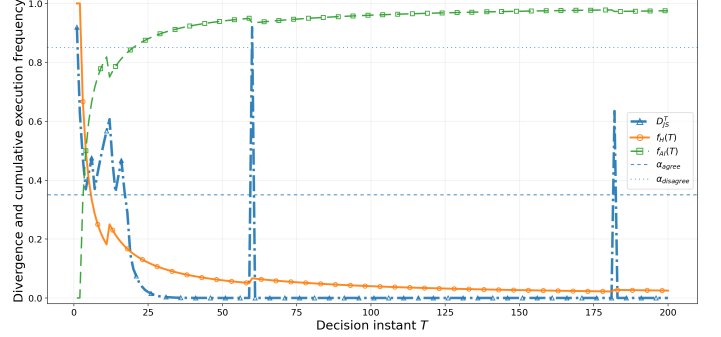
the AI and Human execution frequencies are 97.0% and 3.0. Agreement, contextual, and disagreement occupy 88.5%, 10.5%, and 1.0% of the path. The slight decay therefore does not materially alter the exploratory dynamics at $H = 200$.

Under exogenous generation with $p_A = 0.99$, the AI again executes 97.5% of the decisions and agreement reaches 90.5%. The two rare visits to state B generate the isolated divergence spikes, and the terminal state- B divergence remains 0.4890, despite a near-zero terminal divergence in the frequent state. At this horizon, the contrast between the two panels is governed more strongly by state coverage than by the modest reduction in the global learning rate.

ii. Divergence and execution under global diminishing- η learning



(a) Endogenous generation ($r_s = F$); $\eta_T = 0.15/(1 + 0.0005T)$, $\alpha_{\text{agree}} = 0.35$, $\alpha_{\text{disagree}} = 0.85$.



(b) Exogenous generation ($r_s = T$, $p_A = 0.99$); $\eta_T = 0.15/(1 + 0.0005T)$, $\alpha_{\text{agree}} = 0.35$, $\alpha_{\text{disagree}} = 0.85$.

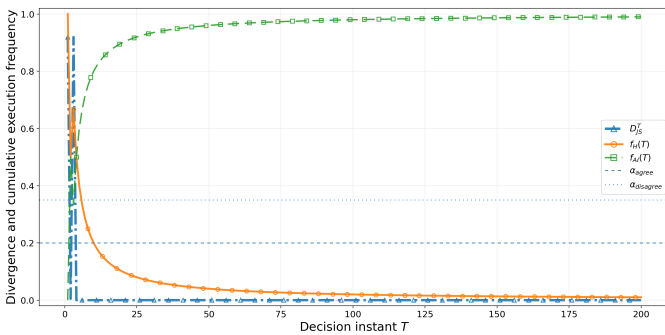
Figure 11: Jensen–Shannon divergence and cumulative execution frequencies at $H = 200$ under global diminishing- η learning. The mild within-horizon decay leaves the trajectories close to their constant-gain counterparts, while extreme exogenous imbalance prevents two-state alignment.

2.3.3 Divergence and execution under exact empirical learning

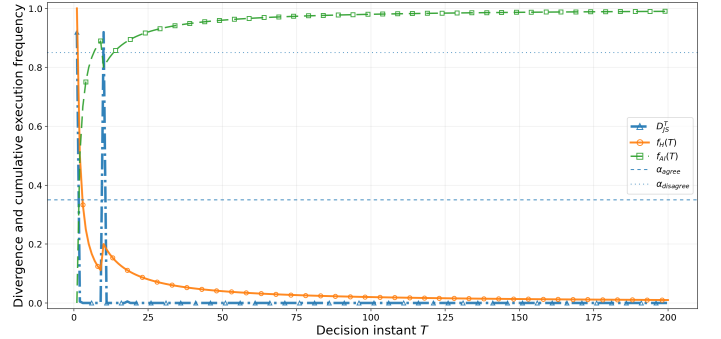
Exact empirical updating produces the sharpest stabilization. In the endogenous run, the divergence becomes identically zero in both states from approximately $T = 5$. The AI executes 99.0% of the decisions, the Human executes 1.0%, and the agreement, contextual, and disagreement frequencies are 99.0%, 0%, and 1.0%. Although state B is visited less frequently than state A , both receive effective observations and finish with exact policy identity.

In the exogenous run, $p_A = 0.10$ yields empirical state frequencies of 9.0% for A and 91.0% for B . The initial state-conditioned spikes disappear after both policy components have been effectively updated, and exact two-state alignment remains stable from approximately $T = 18$. AI execution and agreement again reach 99.0%. The result reflects the local schedule $\eta_{i,T}(s) = 1/[N_i(s, T - 1) + 1]$, whose first effective update has unit gain and whose subsequent updates reproduce the corresponding empirical action frequencies.

iii. Divergence and execution under exact empirical learning



(a) Endogenous generation ($r_s = F$); $\eta_{i,T}(s) = 1/[N_i(s, T - 1) + 1]$, $\alpha_{\text{agree}} = 0.20$, $\alpha_{\text{disagree}} = 0.35$.



(b) Exogenous generation ($r_s = T$, $p_A = 0.10$); $\eta_{i,T}(s) = 1/[N_i(s, T - 1) + 1]$, $\alpha_{\text{agree}} = 0.35$, $\alpha_{\text{disagree}} = 0.85$.

Figure 12: Jensen–Shannon divergence and cumulative execution frequencies at $H = 200$ under exact empirical learning. Once both states receive effective observations, the divergence collapses to zero and the agreement regime dominates the remaining path.

Across the complete $H = 200$ grid, exact empirical learning attains practical and numerical two-state alignment in every evaluated configuration. Under endogenous state generation, practical alignment is reached in 67.50% of constant- η configurations and

62.50% of global diminishing- η configurations. The corresponding exogenous proportions fall to 23.33% and 21.11%, with the failures concentrated at extreme values of p_A . Hence, neither a high cumulative agreement frequency nor AI execution dominance is sufficient evidence of global alignment unless both state-conditioned divergences and state coverage are also examined.

3 Early learning: $H = 1000$

At $H = 1000$, state-wise Human–AI alignment becomes the dominant outcome rather than an isolated short-horizon event. The selected trajectories reach negligible divergence in both states, although some aligned policies continue moving after the Human and AI curves have merged. This horizon is therefore interpreted as an early-learning regime with widespread alignment and residual common-policy adaptation.

3.1 Policy trajectories

For the comparisons below, define

$$D_{JS,\max}^{\text{final}} = \max\{D_{JS,A}^{\text{final}}, D_{JS,B}^{\text{final}}\}.$$

Only the a_0 trajectories are displayed at this horizon. Because the action space is binary, the omitted a_1 curves are exact pointwise complements and add no independent alignment information; this symmetry was documented explicitly at $H = 200$.

3.1.1 Constant- η learning

With endogenous state generation, $\eta = 0.15$ yields practical alignment from approximately $T = 86$ and numerical alignment from $T = 129$. The common policy remains mobile until roughly $T = 532$, showing that Human–AI alignment can precede stationarity of the shared policy. State frequencies are 0.709 in A and 0.291 in B ; the terminal policies satisfy $p_H(a_0 | A) = p_{AI}(a_0 | A) = 1$ and $p_H(a_0 | B) \approx p_{AI}(a_0 | B) \approx 0$, with $D_{JS,\max}^{\text{final}} = 0$. The AI executes 99.1% of the decisions and agreement occupies 95.4% of the path.

The exogenous run uses $\eta = 0.08$ and $p_A = 0.70$, producing balanced enough coverage for two-state learning. Practical and numerical alignment become stable from approximately $T = 100$ and $T = 281$, respectively. Both agents finish at probability 1 in state A and approximately 0.1103 in state B , while the common aligned policy continues to move through much of the horizon. AI execution and agreement both reach 99.2%. Across the full constant- η grid, practical two-state alignment reaches 95.00% under endogenous transitions and 57.92% under exogenous sampling; the remaining failures are concentrated at extreme values of p_A .

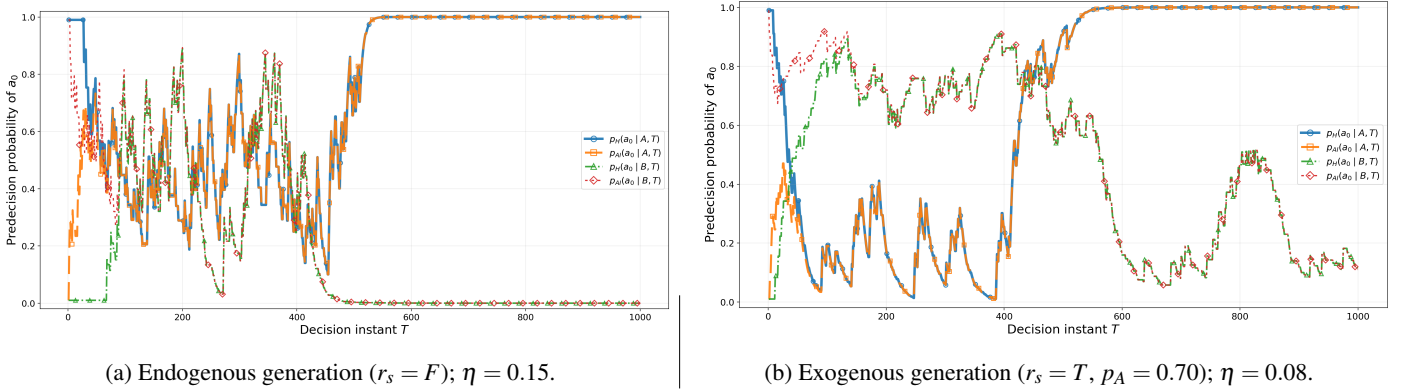
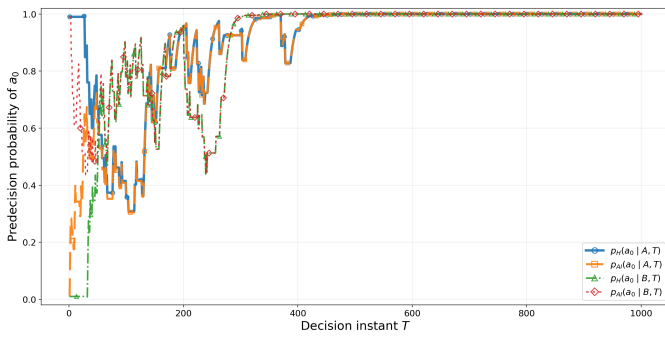


Figure 13: Predecision Human and AI probabilities assigned to action a_0 at $H = 1000$ under constant- η learning. Both runs reach state-wise alignment, although the common aligned policy remains mobile after the divergence has contracted.

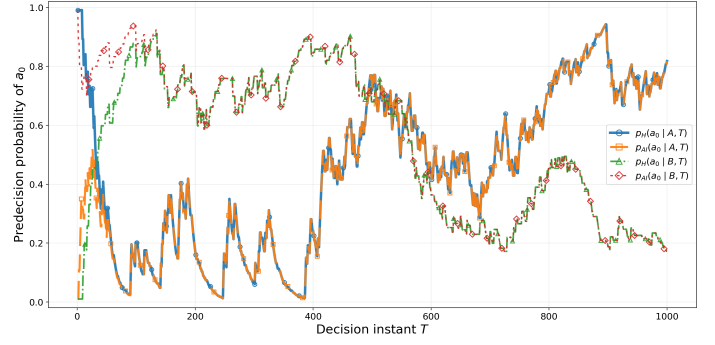
3.1.2 Global diminishing- η learning

The endogenous schedule $\eta_T = 0.15/(1 + 0.001T)$ decreases to approximately 0.075 by the end of the horizon. Practical alignment becomes stable around $T = 47$, numerical alignment around $T = 140$, and the common policy approaches its terminal values around $T = 419$. State frequencies are 0.472 in A and 0.528 in B ; both terminal policies assign probability 1 to a_0 , with $D_{JS,\max}^{\text{final}} = 8.08 \times 10^{-28}$. AI execution reaches 99.4%, while agreement reaches 97.7%.

For exogenous generation, $\eta_T = 0.10/(1 + 0.001T)$ and $p_A = 0.70$. Practical and numerical alignment are stable from approximately $T = 89$ and $T = 226$, respectively. The aligned policy continues to drift smoothly and ends near 0.8269 in state A and 0.1718 in state B . AI execution and agreement both reach 99.4%. In the complete grid, practical alignment reaches 92.50% under endogenous transitions and 52.22% under exogenous sampling, again with the strongest finite-horizon limitations at extreme state probabilities.



(a) Endogenous generation ($r_s = F$); $\eta_T = 0.15/(1 + 0.001T)$.



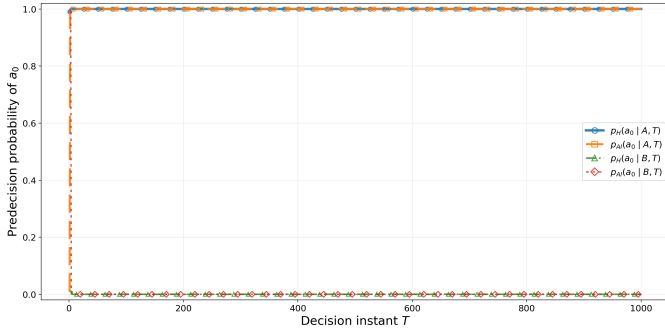
(b) Exogenous generation ($r_s = T$, $p_A = 0.70$); $\eta_T = 0.10/(1 + 0.001T)$.

Figure 14: Predecision Human and AI probabilities assigned to action a_0 at $H = 1000$ under global diminishing- η learning. The schedule contracts policy discrepancy while permitting residual motion of the common aligned policy.

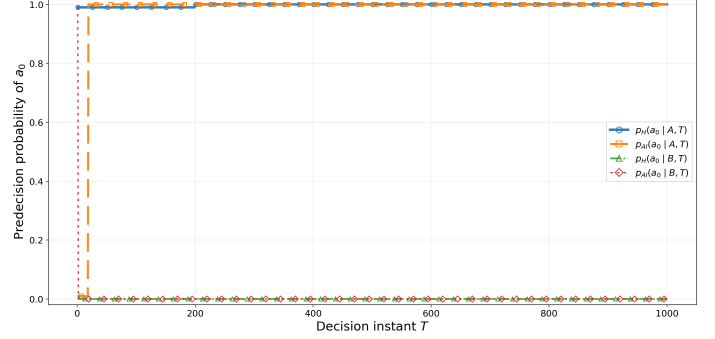
3.1.3 Exact empirical learning

Under exact empirical learning, $\eta_{i,T}(s) = 1/[N_i(s, T-1) + 1]$, so the first effective update of each agent-state component has unit gain. In the endogenous run, practical alignment is stable from approximately $T = 4$, numerical alignment from $T = 6$, and both policies rapidly lock into $p(a_0 | A) = 1$ and $p(a_0 | B) = 0$. AI execution and agreement each reach 99.8%.

The exogenous run deliberately uses $p_A = 0.01$, so state A is observed only eight times. Practical alignment follows the first rare state observation and complete numerical identity follows the second. The final policies again satisfy $p(a_0 | A) = 1$ and $p(a_0 | B) = 0$, with AI execution and agreement at 99.8%. Every exact-empirical configuration in the full grid reaches practical and numerical two-state alignment, including the extreme state probabilities. The visible delay therefore reflects rare-state arrival time rather than slow adaptation after observation.



(a) Endogenous generation ($r_s = F$).



(b) Exogenous generation ($r_s = T$, $p_A = 0.01$).

Figure 15: Predecision Human and AI probabilities assigned to action a_0 at $H = 1000$ under exact empirical learning, with $\eta_{i,T}(s) = 1/[N_i(s, T-1) + 1]$. The exogenous transition records the arrival of information about the rarely sampled state.

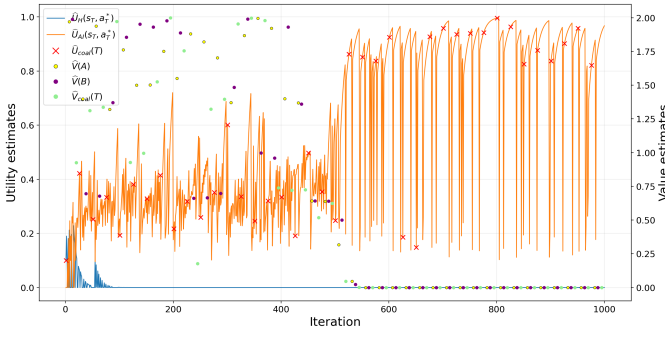
3.2 Utility-credit traces and value iterates

3.2.1 Persistent adaptation and rare-state empirical lock-in

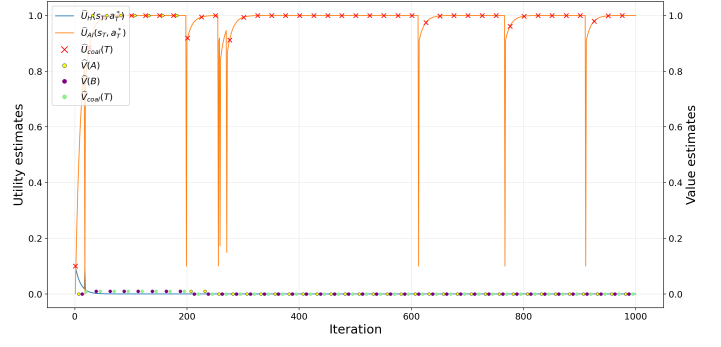
The two displayed diagnostics were selected to contrast persistent finite-time adaptation with exact empirical lock-in. In the constant- η , endogenous run, AI execution reaches 99.1%. The final realized Human credit is numerically zero, whereas the AI and coalition credits are approximately 0.9672. The terminal value iterates are also numerically negligible. The repeated oscillations reflect changes in the realized state-action table entry and the nonvanishing update gain, not resets of the stored tables.

The exact-empirical exogenous run uses $p_A = 0.01$. Its long sequence of state- B observations produces a final AI and coalition credit close to 1, while the isolated downward excursions coincide with rare visits to state A . AI execution and agreement both reach 99.8%, and the terminal value iterates become numerically negligible after the local empirical policies lock in.

The four nondisplayed utility plots provide complementary but visually redundant diagnostics. The constant exogenous run ends with AI and coalition credit 0.7570 and terminal values $\hat{V}(A) = 0.1557$, $\hat{V}(B) = 0.2413$. The global diminishing endogenous and exogenous runs end with AI credits 0.8926 and 0.6825, respectively; their different value scales mainly reflect $\gamma = 0.50$ versus $\gamma = 0.90$. The exact empirical endogenous run ends with AI credit 0.9106 and negligible terminal values. These traces are finite-time diagnostics; the primary convergence evidence remains state-conditioned policy alignment.



(a) Constant η , endogenous generation ($r_s = F$); $\eta = 0.15$, $\beta = 0.10$, $\gamma = 0.50$.



(b) Exact empirical, exogenous generation ($r_s = T$, $p_A = 0.01$); $\beta = 0.10$, $\gamma = 0.50$.

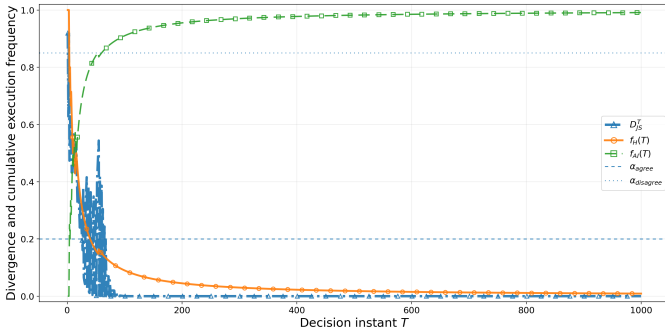
Figure 16: Selected utility-credit traces and finite-time value iterates at $H = 1000$. The left panel shows persistent adaptation under nonvanishing gain; the right panel isolates rare-state arrivals under exact empirical learning.

3.3 Jensen–Shannon divergence and execution frequencies

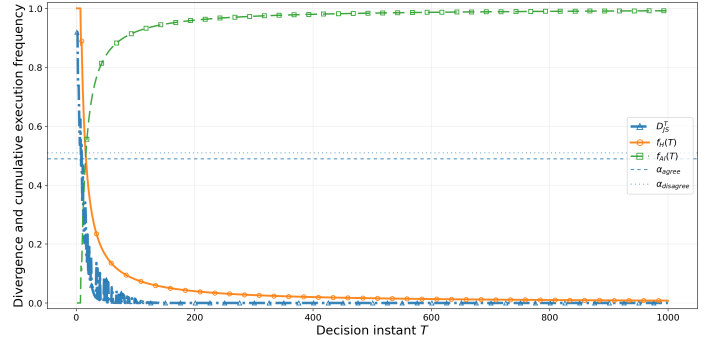
The divergence curves report the predecision D_{JS}^T in the state realized at instant T , together with cumulative Human and AI execution frequencies. Their interpretation therefore combines the plotted path with state-conditioned terminal divergence and empirical state coverage.

3.3.1 Constant- η learning

In the endogenous run, the divergence crosses the execution thresholds during the initial transient and then collapses to zero. AI and Human execution frequencies are 99.1% and 0.9%, while agreement, contextual, and disagreement occupy 95.4%, 4.4%, and 0.2% of the path. The exogenous run contracts more smoothly: AI execution and agreement both reach 99.2%, contextual occupation is zero, and disagreement accounts for 0.8%. Both state-conditioned terminal divergences are zero. Moderate exogenous coverage therefore removes the rare-state failure observed at $H = 200$, although the aligned policy can continue moving after divergence vanishes.



(a) Endogenous generation ($r_s = F$); $\eta = 0.15$, $\alpha_{\text{agree}} = 0.20$, $\alpha_{\text{disagree}} = 0.85$.

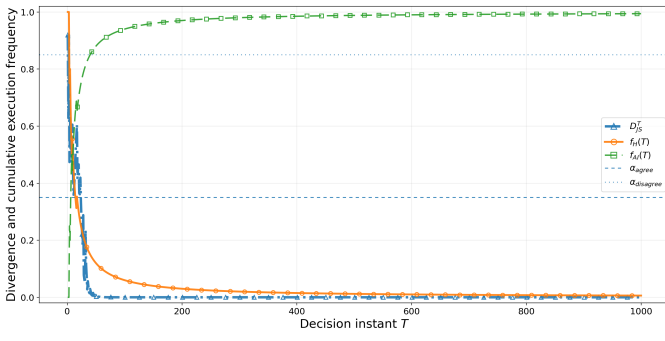


(b) Exogenous generation ($r_s = T$, $p_A = 0.70$); $\eta = 0.08$, $\alpha_{\text{agree}} = 0.49$, $\alpha_{\text{disagree}} = 0.51$.

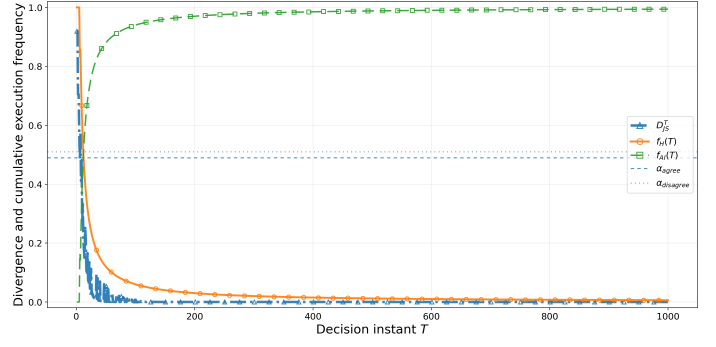
Figure 17: Jensen–Shannon divergence and cumulative execution frequencies at $H = 1000$ under constant- η learning.

3.3.2 Global diminishing- η learning

The endogenous divergence contracts rapidly and remains negligible after the early transient. AI execution reaches 99.4%; agreement, contextual, and disagreement frequencies are 97.7%, 2.1%, and 0.2%. The exogenous run also assigns 99.4% of execution to the AI, with agreement and disagreement frequencies of 99.4% and 0.6%, and no contextual occupation. In both cases, terminal state-wise divergence is zero or numerically negligible. Relative to constant gain, the endogenous schedule shows an earlier and cleaner contraction, while the exogenous path again separates Human–AI alignment from stationarity of the common policy.



(a) Endogenous generation ($r_s = F$); $\eta_T = 0.15/(1 + 0.001T)$, $\alpha_{\text{agree}} = 0.35$, $\alpha_{\text{disagree}} = 0.85$.

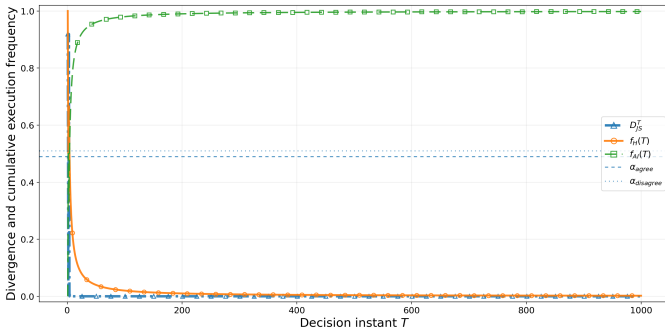


(b) Exogenous generation ($r_s = T$, $p_A = 0.70$); $\eta_T = 0.10/(1 + 0.001T)$, $\alpha_{\text{agree}} = 0.49$, $\alpha_{\text{disagree}} = 0.51$.

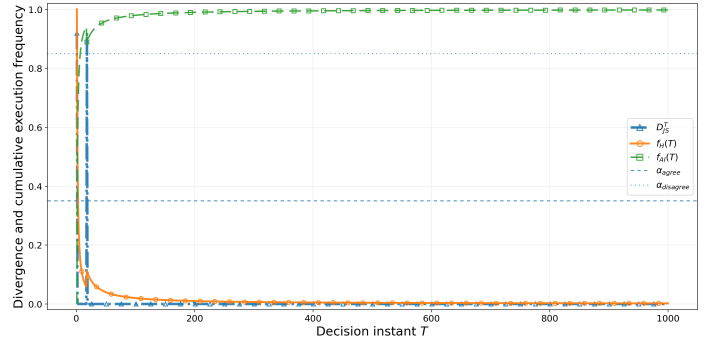
Figure 18: Jensen–Shannon divergence and cumulative execution frequencies at $H = 1000$ under global diminishing- η learning.

3.3.3 Exact empirical learning

Exact empirical updating yields the shortest divergence transient. In the endogenous run, both state-conditioned policies are numerically identical from approximately $T = 6$; AI execution and agreement equal 99.8%, with the remaining 0.2% assigned to disagreement. In the exogenous run, divergence is zero in the frequently visited state and spikes only when the rare state first appears. Practical alignment follows the first rare-state observation, and complete numerical identity follows the second. AI execution and agreement again reach 99.8%. The apparent delay is therefore caused by rare-state arrival time, not by slow updating after observation.



(a) Endogenous generation ($r_s = F$); $\eta_{i,T}(s) = 1/[N_i(s, T-1) + 1]$, $\alpha_{\text{agree}} = 0.49$, $\alpha_{\text{disagree}} = 0.51$.



(b) Exogenous generation ($r_s = T$, $p_A = 0.01$); $\eta_{i,T}(s) = 1/[N_i(s, T-1) + 1]$, $\alpha_{\text{agree}} = 0.35$, $\alpha_{\text{disagree}} = 0.85$.

Figure 19: Jensen–Shannon divergence and cumulative execution frequencies at $H = 1000$ under exact empirical learning. Rare-state spikes disappear immediately after the corresponding effective local updates.

Across the complete $H = 1000$ grid, practical two-state alignment is reached by 95.00% of constant- η and 92.50% of global diminishing- η configurations under endogenous transitions. The corresponding exogenous proportions are 57.92% and 52.22%, a substantial improvement over $H = 200$, but still sensitive to extreme state probabilities. Exact empirical learning reaches practical and numerical identity throughout both grids. Hence, $H = 1000$ marks the transition from exploratory alignment to widespread regime stabilization, while preserving a clear distinction between Human–AI policy agreement and stationarity of the common aligned policy.

4 Emerging stabilisation: $H = 5000$

At $H = 5000$, state-wise Human–AI alignment is the prevailing finite-horizon outcome. The selected trajectories show both robust contraction of policy discrepancy and a distinction that is central to the equilibrium interpretation of Part I: the Human and AI policies can become indistinguishable while their common policy continues to evolve. This horizon is therefore treated as an emerging-stabilisation regime, with strong alignment but schedule- and coverage-dependent residual motion.

4.1 Policy trajectories

For the comparisons below, define

$$D_{JS,\max}^{\text{final}} = \max\{D_{JS,A}^{\text{final}}, D_{JS,B}^{\text{final}}\}.$$

Only the a_0 trajectories are displayed. Since the action space is binary, the omitted a_1 trajectories are exact pointwise complements and contain no additional alignment information; their symmetry was documented explicitly at $H = 200$.

4.1.1 Constant- η learning

With endogenous state generation and $\eta = 0.05$, practical alignment is stable from approximately $T = 87$, numerical alignment from $T = 331$, and the Human and AI curves are indistinguishable to extreme precision from about $T = 798$. The terminal common policy assigns probability 0.6771 to a_0 in state A and probability 1 in state B , with $D_{JS,\max}^{\text{final}} = 2.45 \times 10^{-91}$. The aligned policy continues moving during the horizon because the gain remains nonvanishing. AI execution and agreement both reach 99.76%.

The exogenous run uses the same gain with $p_A = 0.30$, which gives both states substantial coverage. Practical and numerical alignment are stable from approximately $T = 349$ and $T = 659$, respectively, and both policies ultimately assign probability 1 to a_0 in both states. AI execution reaches 99.68%; agreement, contextual, and disagreement frequencies are 96.38%, 3.58%, and 0.04%. Across the complete grid, practical two-state alignment is obtained in 100% of the endogenous constant-gain configurations and 82.92% of the exogenous configurations. Under exogenous sampling, the corresponding proportion is 100% at $p_A = 0.30$, but falls to 50% at $p_A = 0.99$, confirming that state coverage remains a finite-horizon constraint.

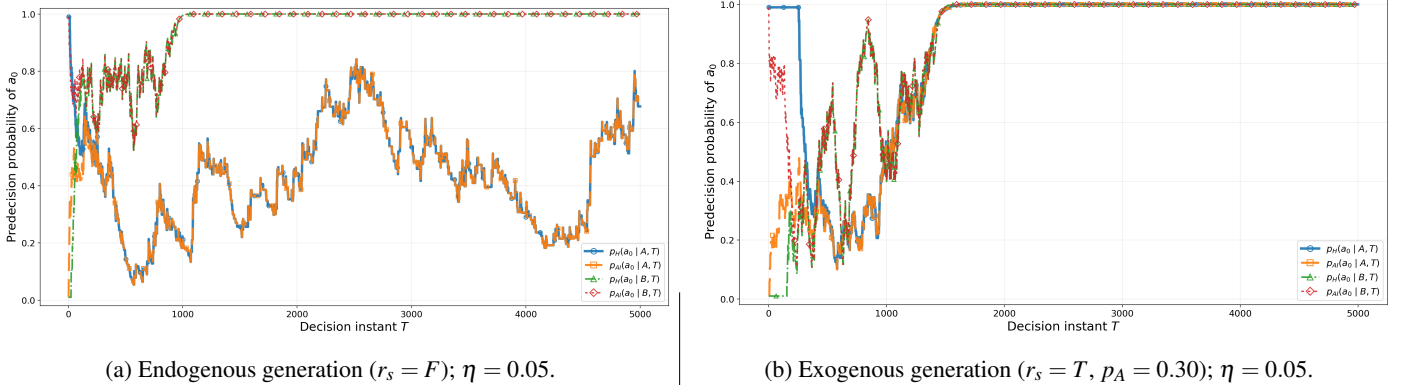


Figure 20: Predecision Human and AI probabilities assigned to action a_0 at $H = 5000$ under constant- η learning. The panels show robust state-wise alignment while also distinguishing alignment from freeze-out of the common policy.

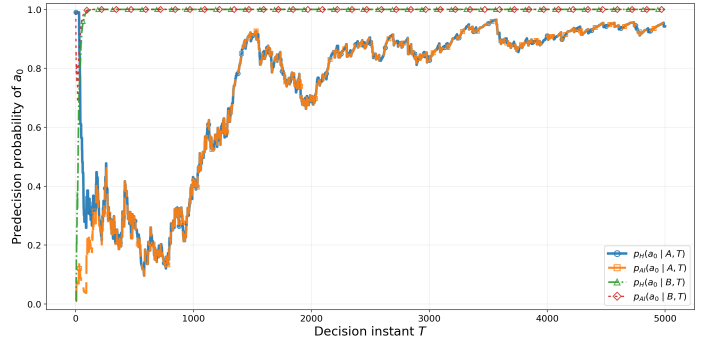
4.1.2 Global diminishing- η learning

The endogenous schedule $\eta_T = 0.05/(1 + 0.001T)$ decreases to approximately 0.0083 by the end of the horizon. Practical and numerical alignment are stable from approximately $T = 313$ and $T = 571$, respectively. Both agents finish at $p(a_0 | A) = 0.4984$ and $p(a_0 | B) = 0.4440$, with zero terminal divergence in both states. AI execution reaches 99.50%, while agreement and contextual occupation reach 96.42% and 3.34%. The trajectory is especially informative because policy alignment occurs well before the common policy has ceased to move.

Under exogenous generation, $\eta_T = 0.08/(1 + 0.001T)$ and $p_A = 0.30$. Practical alignment is stable from approximately $T = 126$, numerical alignment from $T = 334$, and the two policies become indistinguishable to extreme precision from about $T = 784$. Their terminal probabilities are 0.9434 in state A and 1 in state B . AI execution is 99.94% and agreement occupies 99.86% of the path. In the full grid, practical alignment reaches 95.00% for endogenous transitions and 61.39% for exogenous sampling. The exogenous rate is 94.17% at $p_A = 0.30$, but only 10.00% at $p_A = 0.99$, because late observations of the rare state receive a small globally decayed gain.



(a) Endogenous generation ($r_s = F$); $\eta_T = 0.05/(1 + 0.001T)$.



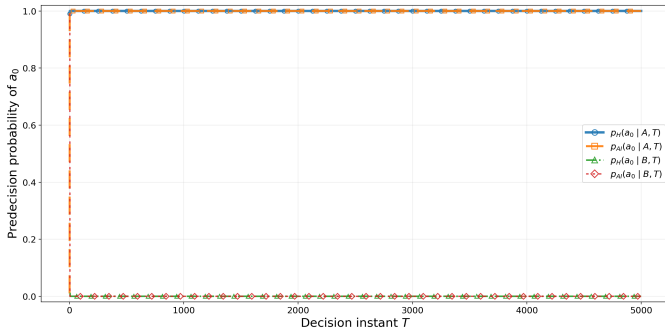
(b) Exogenous generation ($r_s = T$, $p_A = 0.30$); $\eta_T = 0.08/(1 + 0.001T)$.

Figure 21: Predecision Human and AI probabilities assigned to action a_0 at $H = 5000$ under global diminishing- η learning. The aligned policies retain smooth residual motion, particularly in the endogenous run.

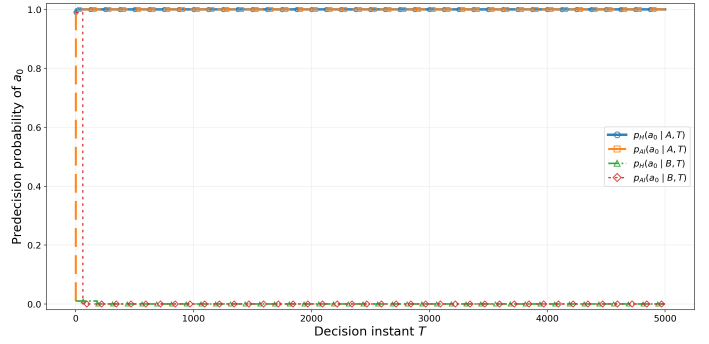
4.1.3 Exact empirical learning

Under exact empirical learning, $\eta_{i,T}(s) = 1/[N_i(s, T-1) + 1]$, and each agent-state component adapts according to its own effective observation count. In the endogenous run, practical alignment is stable from approximately $T = 3$ and numerical alignment from $T = 5$. Both policies rapidly lock into $p(a_0 | A) = 1$ and $p(a_0 | B) = 0$. AI execution and agreement each reach 99.96%.

The exogenous run deliberately uses the extreme probability $p_A = 0.99$. State B is visited only 59 times, with its first two informative arrivals occurring around $T = 60$ and $T = 182$. Nevertheless, local empirical updating produces exact identity in both states: practical alignment becomes stable from the first rare-state arrival and numerical alignment from the second. AI execution and agreement again reach 99.96%. Exact empirical learning achieves practical and numerical two-state alignment throughout the entire endogenous and exogenous grids, including all extreme values of p_A .



(a) Endogenous generation ($r_s = F$).



(b) Exogenous generation ($r_s = T$, $p_A = 0.99$).

Figure 22: Predecision Human and AI probabilities assigned to action a_0 at $H = 5000$ under exact empirical learning, with $\eta_{i,T}(s) = 1/[N_i(s, T-1) + 1]$. The exogenous panel shows that local update counts preserve adaptation to a rarely observed state.

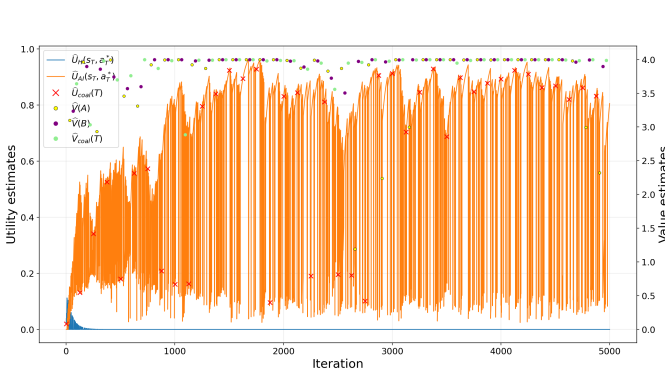
4.2 Utility-credit traces and value iterates

The utility-credit traces and finite-time Bellman iterates are path-dependent computational diagnostics. Their vertical levels are not directly comparable across panels because the selected values of β and γ differ. Abrupt changes may also reflect a change in the realized state-action table entry rather than a reset of the underlying tables.

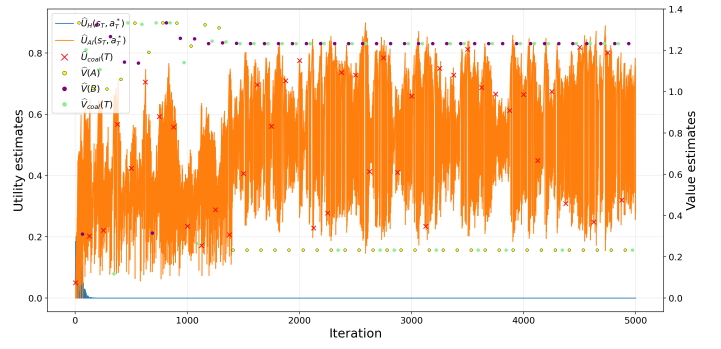
4.2.1 Constant- η learning

In the endogenous run, AI execution reaches 99.76%. The final realized Human credit is zero, while the AI and coalition credits are 0.8059. With $\gamma = 0.75$, the terminal value iterates are $\hat{V}(A) = 3.9930$ and $\hat{V}(B) = 3.9984$. The dense oscillation in the realized traces reflects continuing movement among state-action entries under nonvanishing gain, despite almost complete Human-AI alignment.

The exogenous run uses $\beta = 0.05$, $\gamma = 0.25$, and $p_A = 0.30$. Its final AI and coalition credit is 0.2811, while $\hat{V}(A) = 0.2333$ and $\hat{V}(B) = 1.2333$. AI execution is 99.68%. The lower value scale is primarily induced by the smaller discount factor and should not be interpreted as weaker policy alignment.



(a) Endogenous generation ($r_s = F$); $\eta = 0.05$, $\beta = 0.02$, $\gamma = 0.75$.



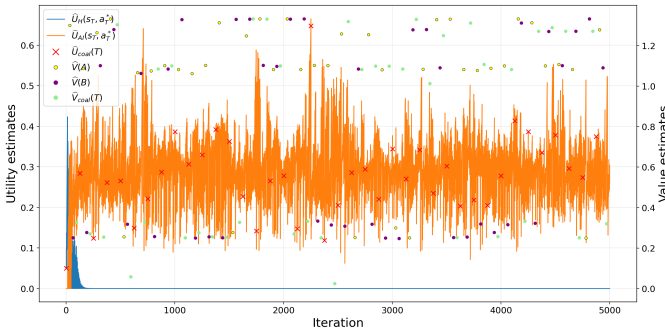
(b) Exogenous generation ($r_s = T$, $p_A = 0.30$); $\eta = 0.05$, $\beta = 0.05$, $\gamma = 0.25$.

Figure 23: Utility-credit traces and finite-time value iterates at $H = 5000$ under constant- η learning. Persistent trace variation coexists with state-wise policy alignment.

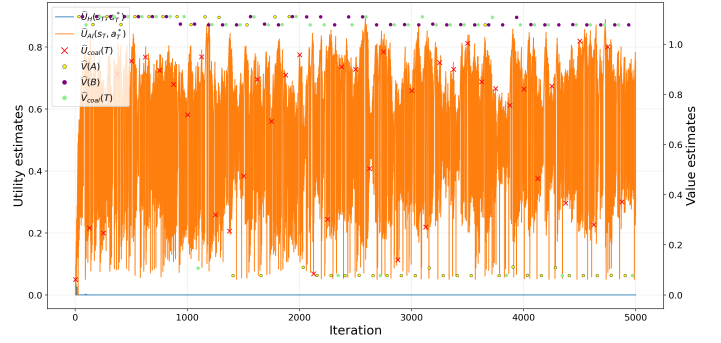
4.2.2 Global diminishing- η learning

The endogenous global-decay run ends with AI and coalition credit 0.2610, and with value iterates 0.2516 in state A and 0.2719 in state B . AI execution reaches 99.50%. The credit series remains variable because the realized state-action pair changes, whereas the value scale remains compact under $\gamma = 0.25$.

For the exogenous run, the final AI and coalition credit is 0.2541. The value iterates end at 0.0790 in state A and 1.0810 in state B , with the realized coalition value ending at the state A level. AI execution reaches 99.94%. The combination of $p_A = 0.30$ and a diminishing gain produces a clean alignment path, while $\gamma = 0.10$ keeps the value scale local to the most recent structural rewards.



(a) Endogenous generation ($r_s = F$); $\eta_T = 0.05/(1 + 0.001T)$, $\beta = 0.05$, $\gamma = 0.25$.



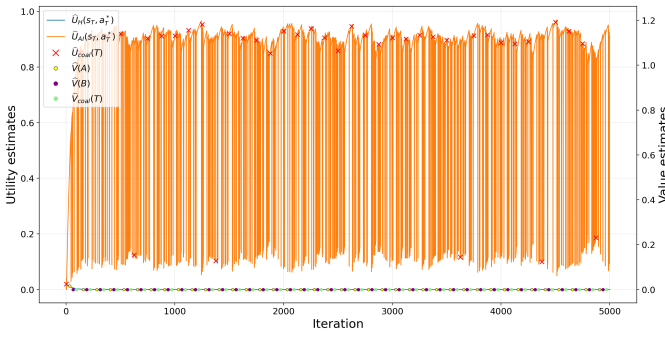
(b) Exogenous generation ($r_s = T$, $p_A = 0.30$); $\eta_T = 0.08/(1 + 0.001T)$, $\beta = 0.05$, $\gamma = 0.10$.

Figure 24: Utility-credit traces and finite-time value iterates at $H = 5000$ under global diminishing- η learning. The panels illustrate aligned policies with continuing path-dependent credit variation.

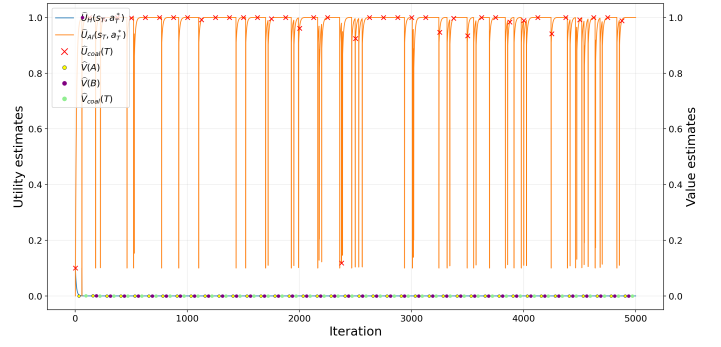
4.2.3 Exact empirical learning

In the endogenous exact-empirical run, AI execution reaches 99.96% and the final AI and coalition credit is 0.9102. Both value iterates are numerically zero after the policy has locked into its empirical action pattern. This does not define convergence by itself; it is a secondary consequence of the realized structural rewards and the selected action sequence.

The exogenous run with $p_A = 0.99$ ends with AI and coalition credit numerically equal to 1. Its occasional downward excursions identify rare state B visits, whereas the value iterates are numerically negligible at the end of the horizon. The trace therefore complements the policy plot by revealing the sparse arrival pattern that the local empirical learning rule successfully incorporates.



(a) Endogenous generation ($r_s = F$); $\beta = 0.02$, $\gamma = 0.50$.



(b) Exogenous generation ($r_s = T$, $p_A = 0.99$); $\beta = 0.10$, $\gamma = 0.10$.

Figure 25: Utility-credit traces and finite-time value iterates at $H = 5000$ under exact empirical learning. The exogenous panel makes the rare-state arrival structure visible even after policy identity has been established.

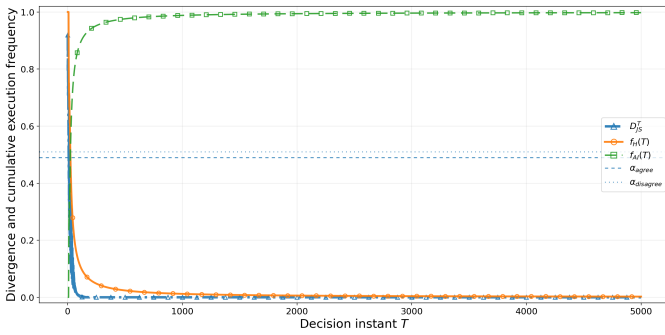
4.3 Jensen–Shannon divergence and execution frequencies

The divergence plots report the predecision D_{JS}^T in the state realized at instant T , together with cumulative Human and AI execution frequencies. Their interpretation is combined with the state-conditioned terminal divergences and the occupation frequencies of the three delegation regions.

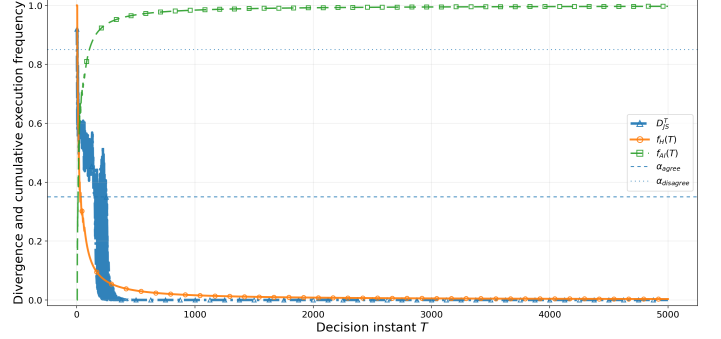
4.3.1 Constant- η learning

In the endogenous run, divergence contracts rapidly and remains negligible after the initial transient. AI and Human execution frequencies are 99.76% and 0.24%; agreement and disagreement occupy 99.76% and 0.24%, with no contextual decisions. The zero terminal divergence in state A and the 2.45×10^{-91} value in state B confirm two-state alignment.

The exogenous run exhibits a wider early divergence band before both state-conditioned values reach zero. AI execution is 99.68%, while agreement, contextual, and disagreement frequencies are 96.38%, 3.58%, and 0.04%. The contextual occupation records the longer transitional approach to alignment; it does not persist once the policies have merged.



(a) Endogenous generation ($r_s = F$); $\eta = 0.05$, $\alpha_{\text{agree}} = 0.49$, $\alpha_{\text{disagree}} = 0.51$.



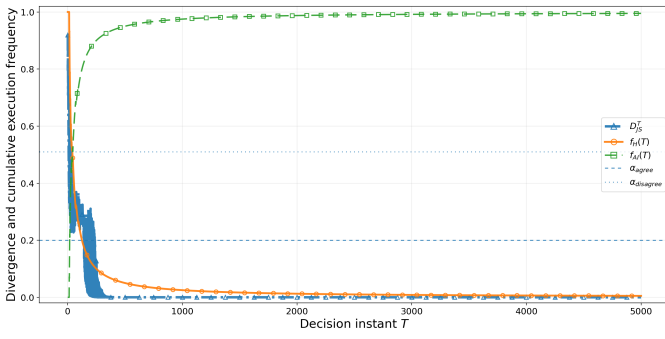
(b) Exogenous generation ($r_s = T$, $p_A = 0.30$); $\eta = 0.05$, $\alpha_{\text{agree}} = 0.35$, $\alpha_{\text{disagree}} = 0.85$.

Figure 26: Jensen–Shannon divergence and cumulative execution frequencies at $H = 5000$ under constant- η learning.

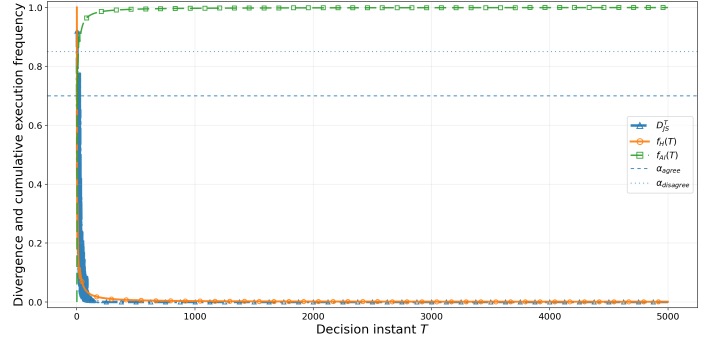
4.3.2 Global diminishing- η learning

The endogenous divergence contracts more gradually than in the selected constant-gain run, but both state-conditioned terminal values are zero. AI execution reaches 99.50%; agreement, contextual, and disagreement frequencies are 96.42%, 3.34%, and 0.24%. The residual policy motion visible in the policy panel is therefore movement of an already aligned pair rather than renewed divergence.

The exogenous run gives the cleanest divergence contraction at this horizon. AI execution reaches 99.94%, while agreement, contextual, and disagreement occupy 99.86%, 0.10%, and 0.04%. Terminal divergence is zero in state A and 6.89×10^{-46} in state B . Moderate state coverage and the selected thresholds yield a short transient followed by stable AI-dominant agreement.



(a) Endogenous generation ($r_s = F$); $\eta_T = 0.05/(1 + 0.001T)$, $\alpha_{\text{agree}} = 0.20$, $\alpha_{\text{disagree}} = 0.51$.



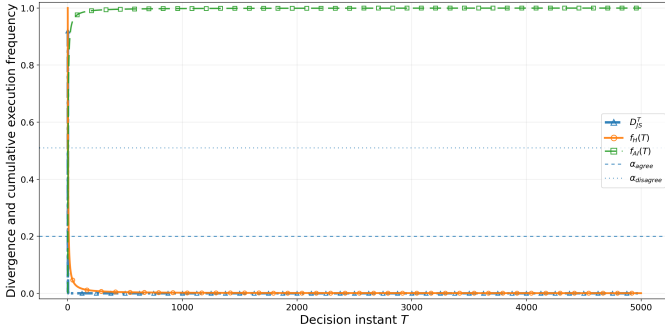
(b) Exogenous generation ($r_s = T$, $p_A = 0.30$); $\eta_T = 0.08/(1 + 0.001T)$, $\alpha_{\text{agree}} = 0.70$, $\alpha_{\text{disagree}} = 0.85$.

Figure 27: Jensen–Shannon divergence and cumulative execution frequencies at $H = 5000$ under global diminishing- η learning.

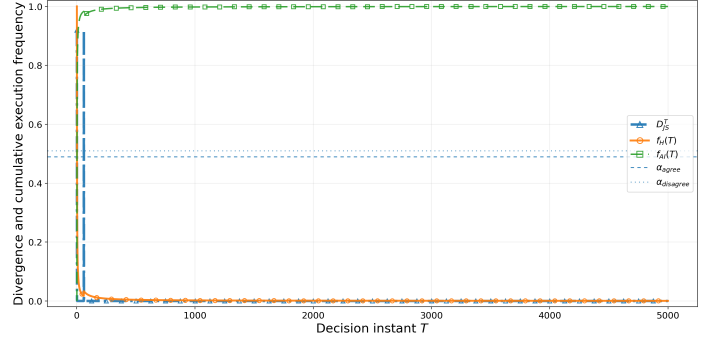
4.3.3 Exact empirical learning

In both exact-empirical runs, the divergence spikes are confined to the first effective observations of the relevant state–agent components. The endogenous run reaches agreement on 99.96% of decisions, with 0.04% disagreement and no contextual occupation. AI execution is also 99.96%, and both terminal state-conditioned divergences are zero.

The exogenous run records the same terminal frequencies despite $p_A = 0.99$. The isolated early spikes correspond to the first rare state B observations; once incorporated, they do not generate persistent misalignment. Agreement and AI execution again reach 99.96%, and both terminal divergences are exactly zero. This result distinguishes local empirical adaptation from global time decay under extreme state imbalance.



(a) Endogenous generation ($r_s = F$); $\alpha_{\text{agree}} = 0.20$, $\alpha_{\text{disagree}} = 0.51$.



(b) Exogenous generation ($r_s = T$, $p_A = 0.99$); $\alpha_{\text{agree}} = 0.49$, $\alpha_{\text{disagree}} = 0.51$.

Figure 28: Jensen–Shannon divergence and cumulative execution frequencies at $H = 5000$ under exact empirical learning. The rare-state spikes are transient and disappear after local incorporation.

4.4 Finite-horizon synthesis

The complete grid confirms that $H = 5000$ marks a qualitative shift from early learning to emerging stabilisation. Practical two-state alignment reaches 100% under endogenous constant gain, 95.00% under endogenous global decay, and 100% under exact empirical learning. Under exogenous sampling, the corresponding proportions are 82.92%, 61.39%, and 100%. Balanced and moderately imbalanced state probabilities produce nearly universal alignment, whereas extreme probabilities remain consequential for globally decaying updates.

The displayed trajectories also show that alignment and policy freeze-out are distinct. Constant-gain and Robbins–Monro runs can produce negligible Human–AI divergence while the common policy still moves, whereas exact empirical updating can lock rapidly into a pure empirical pattern. The primary evidence at this horizon is therefore robust alignment of the two agents and stabilization of execution frequencies, not universal pointwise stationarity of every policy trajectory.

5 END

References

- [1] S. A. Sepúlveda-Fontaine and J. M. Amigó, “An Entropy-Based Framework for Hybrid Coalitions in Game Theory—Part I: Human Arbitration,” *Entropy*, vol. 28, no. 4, Art. 473, 2026. <https://doi.org/10.3390/e28040473>.